

ESC-201 Mechanics of Materials Notes

The main idea of this course is to understand how mechanisms and designs perform under various loads and conditions. We will combine force loading from statics and dynamics with material properties to quantify the performance of a design and see how it fails.

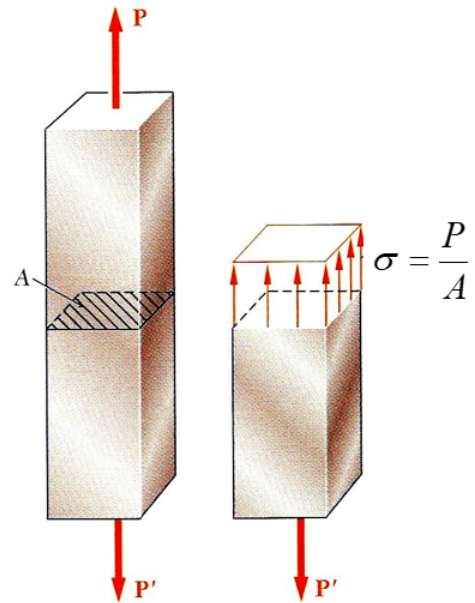
Basic Concepts

Normal Stress

When load is applied normal to a cross-section such as the one below, we can find the normal stress applied by dividing the force by the area of the cross-section. This represents the *average* normal stress, as the actual stress distribution is usually not uniform across the cross-section. It can be thought similarly to pressure, which both represents a load over an area.

$$\sigma_{\text{avg}} = \frac{P}{A}$$

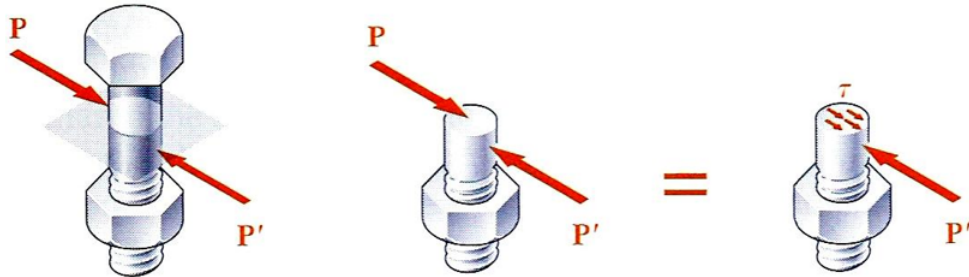
Sign convention: A positive stress corresponds to **tensile** loading while a negative stress corresponds to **compressive** loading.



Shear Stress

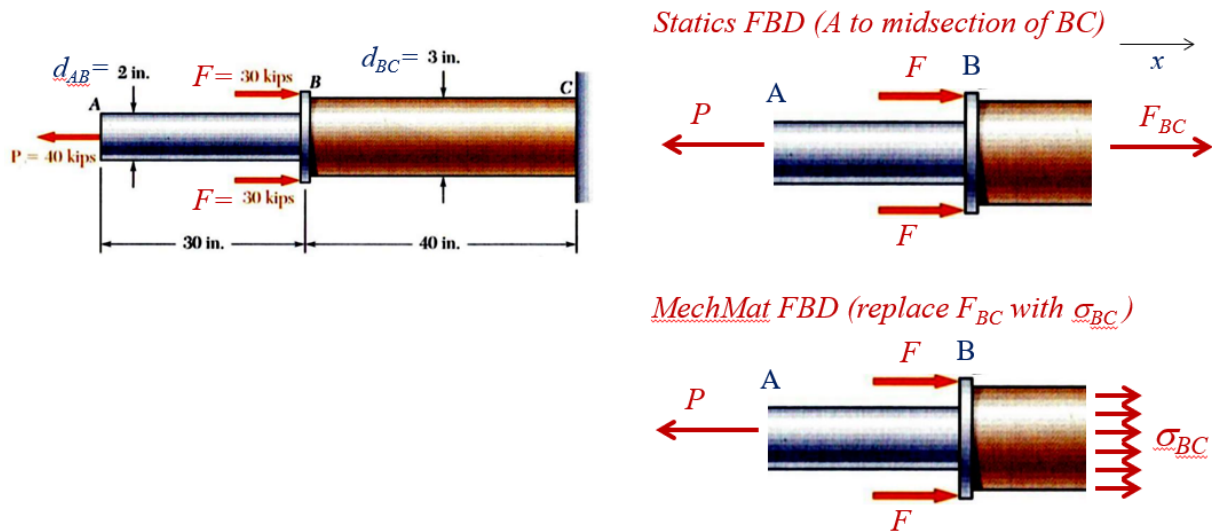
When load is applied parallel or along a cross-section, we can find the shear stress by dividing the force by the area of the cross-section. Similar to normal stress, this represents the *average* shear stress, as the actual stress distribution is usually not uniform across the cross-section.

$$\tau_{\text{avg}} = \frac{P}{A}$$



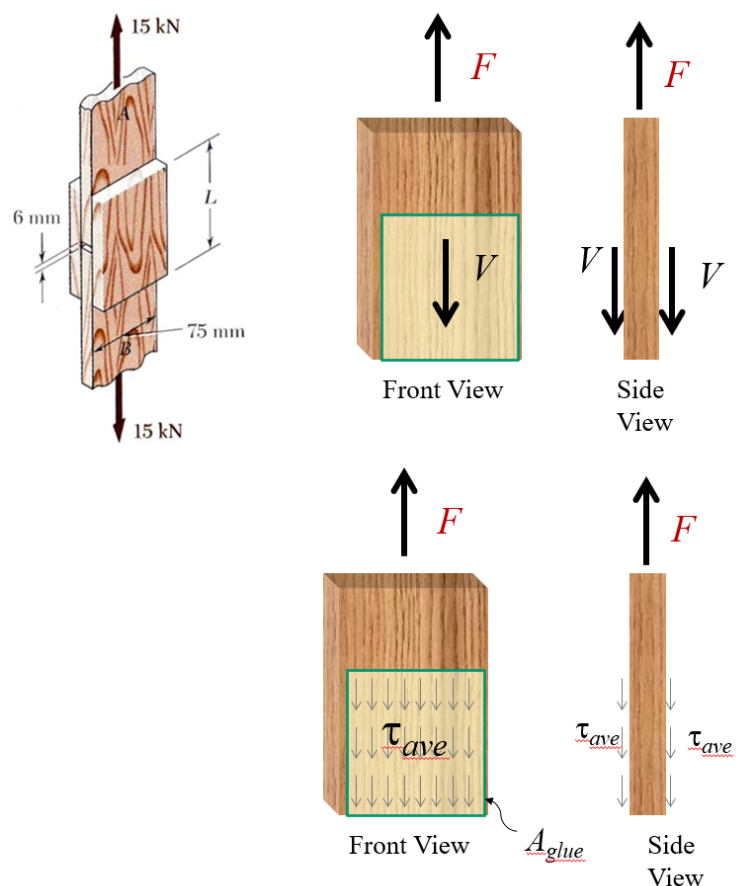
Sign convention: It follows the sign of the applied force. (ie. If the force is in the positive direction, the shear stress is positive.)

Note: It is sometimes useful to think of stress as replacement for forces. For example, we can draw two different FBDs, one showing the forces and one showing the stresses.



This way, it is clear to see what force corresponds to what stress and what area it acts on.

This is representative of our problem-solving approach. Starting from static force balances to solve for the forces, then using those forces to find the stresses, and finally using the stresses to find the strains and deformations.



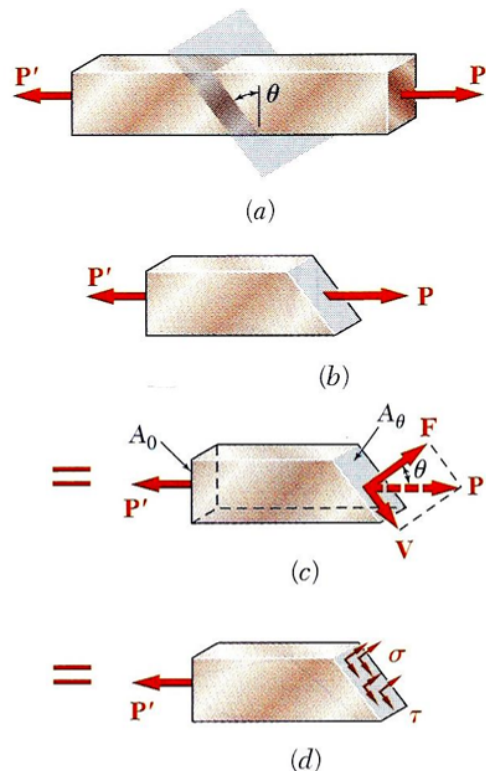
General Stress

When either the load is applied at an angle or the cross-section is not perpendicular to the load, we can always decompose the load into a normal and shear component. Then we can define the normal and shear stresses as before.

$$\sigma = \frac{F}{A_\theta} = \frac{P \cos \theta}{A_0 / \cos \theta} = \frac{P \cos^2 \theta}{A_0}$$

$$\tau = \frac{V}{A_\theta} = \frac{P \sin \theta}{A_0 / \cos \theta} = \frac{P \sin \theta \cos \theta}{A_0}$$

Where A_0 is the area of the cross-section perpendicular to the load/axis and A_θ is the area of the cross-section at an angle θ to the load/axis.



Pinned Joints

When considering a pinned joint and how it fails, there are several ways. Mainly, it depends on whether the load is tensile or compressive. For both cases, the average normal stress at the midsection matters, but it's not usually where failure *first* occurs, or in other words, it's not a critical region (it is usually where stress is max). The following are the critical regions unique to the type of loading:

• Member in Tension

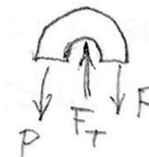
Under tension, failure can occur by the member splitting at the shoulders around the pin where the cross-sectional area is smallest.

$$\sigma_{\text{avg}_{\text{max}}} = \frac{F_T}{A_{\text{min}}} \quad (\text{max average normal stress})$$

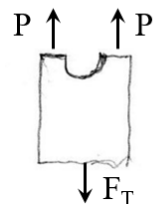
Where A_{min} is the minimum cross-sectional area of the member, which is usually defined by $(w - d)t$ where w is the width of the member, d is the diameter of the pin, and t is the thickness of the member.

Cut member at Pin:

End:



Midsection:

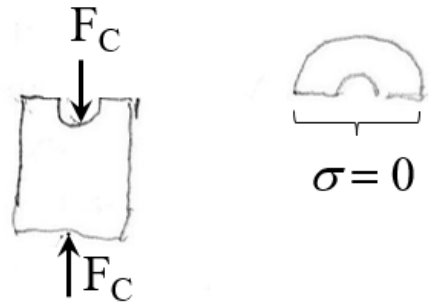


- **Member in Compression**

Under compression, the pin squishes into the member and effectively becomes a solid piece. Thus, the risk isn't at the shoulders but rather at the midsection where it could buckle. Note that there is ZERO stress at the upper region around the hole since there is no load there.

$$\sigma_{\text{avg}} = \frac{F_C}{A_{\text{cross-section}}} \quad (\text{the usual average normal stress})$$

Cut member at Pin:



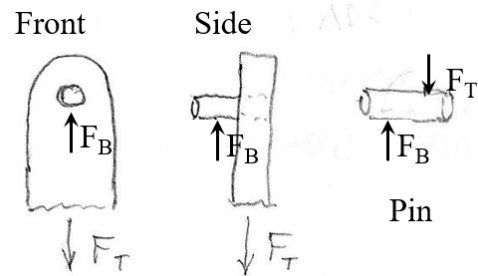
For both cases, we need to also consider the shear stress on the pin and also the bearing stress on the member.

Shear Stress at Pin

The first place failure can occur is that the pin shears. In this case, the pin is in *single shear*. This is the same for both tension and compression.

$$\tau_{\text{avg}} = \frac{F_T}{A_{\text{pin}}}$$

Where A_{pin} is the cross-sectional area of the pin.



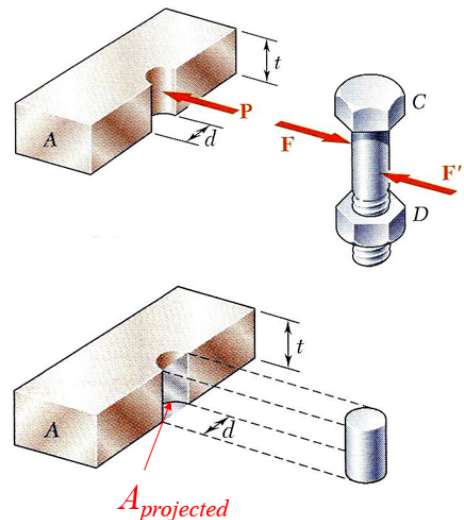
Bearing Stress

Another place failure can occur is that the hole the pin rests upon physically deforms and causes failure. This is called bearing stress, and it is always a compressive stress (the pin pushes into the hole).

It is also an average stress, but it uses the projected area of the bolt onto the hole:

$$\sigma_{\text{avg}} = \frac{P}{A_{\text{projected}}} = \frac{P}{td}$$

Sign convention: Like mentioned before, this is always a compressive stress, so we report it as a positive value.



Factor of Safety and Stresses

Factor of Safety

We use the factory of safety to compensate for using uniform stress assumptions and other variability. It is defined as

$$FS = \frac{\text{Ultimate Load}}{\text{Allowable Load}}$$

Where the ultimate load is the max load at failure and the allowable load is the max design load. If failure is due to stress exceeding the ultimate strength, then

$$FS = \frac{\sigma_{\text{ult}}}{\sigma_{\text{allowable}}}$$

Where σ_{ult} is the ultimate strength of the material and $\sigma_{\text{allowable}}$ is the allowable stress.

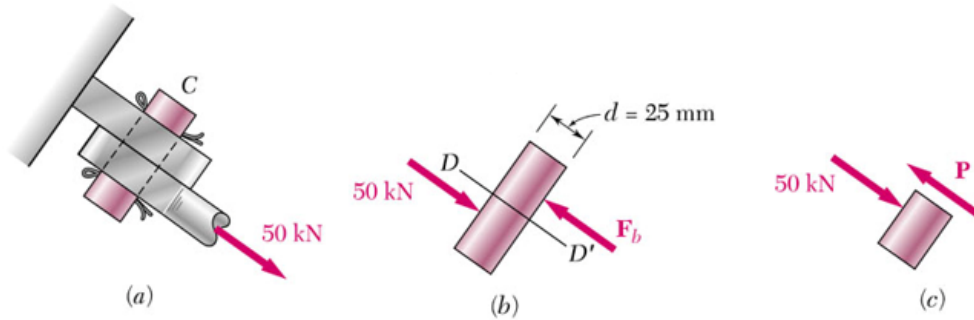
Shear in Pin or Bolt

We've already mentioned single shear as a mode of failure for pinned joints, but here it is again:

Single Shear

Single shear means that there is one member per side of the pin, such that the shear load is *equal* to the member/applied load.

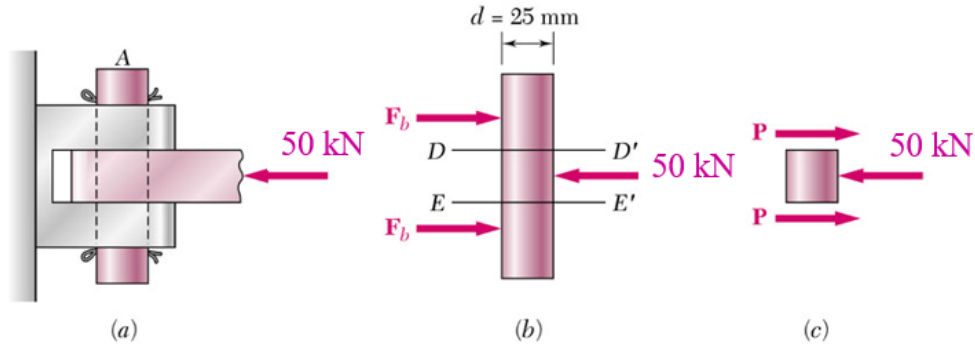
$$\tau_{\text{avg}} = \frac{\text{load}}{\text{pin XC area}}$$



Double Shear

Double shear means shear is carried on both sides of incoming member (clevis joint) such that the load is split between two sides of clevis.

$$\tau_{\text{avg}} = \frac{\text{load}/2}{\text{pin XC area}}$$



Axial Loading and Deformation

This is going to be a long section, we finally introduce strain and deformation, allowing us to find things such as thermal stress, and solve for statically indeterminate problems.

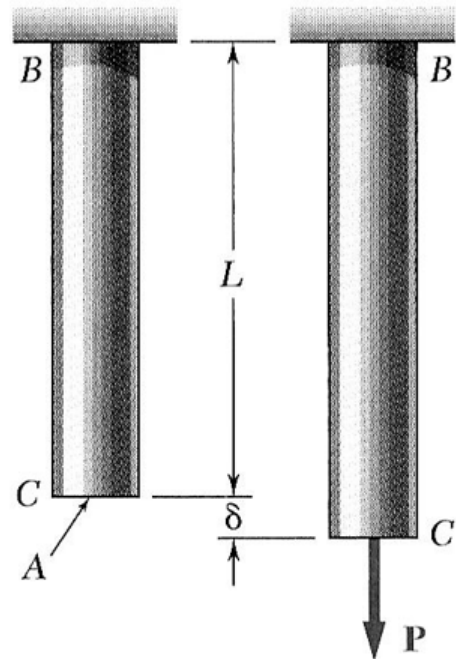
Axial Strain

If deflection δ is defined as the change in length of a member (usually by some load), then we can define the axial strain as the amount of deflection per unit length. It is a dimensionless quantity.

$$\epsilon = \frac{\delta}{L} \quad (\text{uniform strain})$$

For a general member, we can define the limit

$$\epsilon = \lim_{\delta \rightarrow 0} \frac{\Delta \delta}{\Delta L} = \frac{d\delta}{dL} \quad (\text{local strain})$$



Axial Deflection

In the elastic region below the yield stress, we can use Hooke's Law to relate the axial stress to the axial strain using Young's Modulus E .

$$\sigma = E\epsilon$$

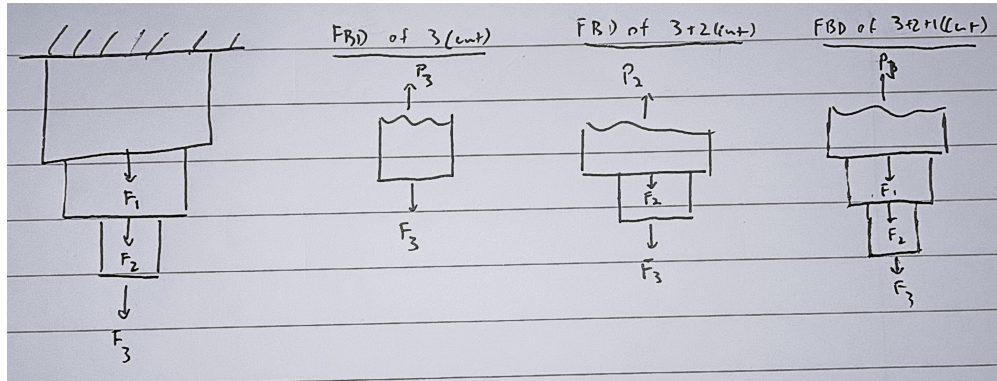
Thus, we can define the **axial deflection** as

$$\delta = \frac{PL}{EA}$$

and **stiffness** of the member as

$$k = \frac{P}{\delta} = \frac{EA}{L}$$

Note that P is an *internal* reaction force, not the applied load, so we must first solve for the internal forces using static force balances.



Thus, we can generalize for non-uniform members by integrating over the length of the member:

$$\delta = \sum \frac{P_i L_i}{E_i A_i} = \int_0^L \frac{P}{EA} dx$$

Statically Indeterminate Problems

The nice thing about having a deflection formula is that we can now apply constraints on the deflection to solve for unknown forces. A useful analogy here is to think of the members as springs. In some cases, the members are in series where the total length is constant ($\delta_1 + \delta_2 = 0$), and in some cases, the members are in parallel where the deflection is constant ($\delta_1 = \delta_2$). We can use these constraints to solve for unknown forces and then find the stresses and strains as usual.